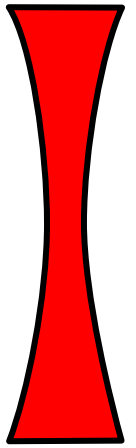
Lens formula – diverging lens

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

f is negative for a diverging lens

v is negative (virtual image)



u = Object distance v = Image distance

Negatives in the lens formula

1. u is always positive
2. f is negative if the lens is diverging
3. v is negative if it's a virtual image

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

The formula in the tables has no negative signs.

Use the formula in the tables.

Example 6

Diverging lenses are used in doors to allow house occupants see who is outside. A man is standing 1 m away from such a lens. The lens has a focal length of 20 cm, How far from the lens will his image appear?

$$\frac{1}{u}+\frac{1}{v}=\frac{1}{f}$$

$$\frac{1}{100}+\frac{1}{v}=\frac{1}{-20} \quad diverging\ lens\ means\ f\ is\ negative$$

$$\frac{1}{v}=-\frac{1}{20}-\frac{1}{100}=\frac{-5-1}{100}$$

$$v=-\frac{100}{6}=-16.6\ cm \qquad negative \Rightarrow virtual$$

Exercise

A diverging lens has a focal length of 20 cm. An object is placed 15 cm from the lens,

(i) calculate how far the image will be from the lens.

(ii) Is the image real or virtual.

Tip: recall that f is negative for diverging lenses

-8.57 cm

Virtual image